1 Overview

The coastal sensor network measures water temperature, salinity and wave height along a forty kilometre shoreline. Each station reports a compact summary every ten minutes and keeps the raw samples for a full week. The network was installed in three phases between early spring and late autumn. Most stations run on solar power and a small battery that lasts about four cloudy days.

This report describes the hardware, the data path from the water to the archive, and the checks that run every night. It also lists the faults we saw during the first winter and the changes we made in response. Readers who only need the data format can skip ahead to the archive section.

2 Hardware

A station consists of a moored buoy, a short radio link and a relay mast on the nearest pier. The buoy carries a temperature probe, a conductivity cell and a pressure sensor for waves. Probes are cleaned by a small wiper that runs once an hour to slow the growth of algae. The relay mast forwards the packets to a gateway in the harbour office over a licensed band.

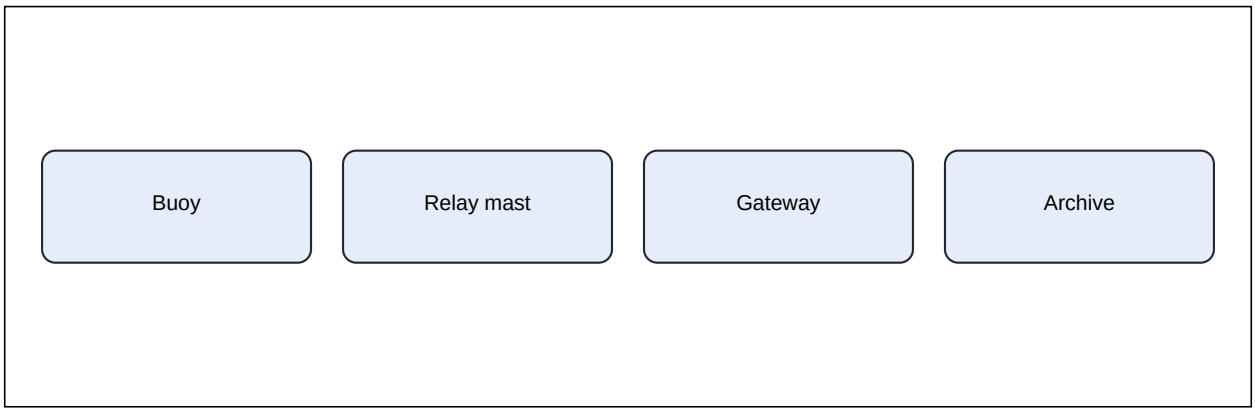

Figure 1: Data path of one station. Packets travel from the water to the archive in four hops.

The gateway buffers packets for up to two days when the uplink to the archive is down. During the storm in November the buffer filled twice and the oldest summaries were dropped. We have since doubled the buffer and added an alarm when it passes eighty percent.

3 Measurements

Temperature readings are calibrated against a reference thermometer before every deployment. Salinity is derived from conductivity and temperature using the standard practical scale. Wave height is estimated from the pressure record over a twenty minute window. All three values are stored with a quality flag that marks suspect or missing samples.

Table 1: Sensor specifications for a standard station.

Sensor	Range	Accuracy	Rate
Temperature	-2 to 35 C	0.05 C	1 Hz
Conductivity	0 to 70 mS/cm	0.1 mS/cm	1 Hz
Pressure	0 to 20 dbar	0.02 dbar	4 Hz
Battery	10 to 14 V	0.1 V	0.1 Hz

The quality flags are set by simple range checks and by a spike detector that compares each sample with its neighbours. A sample that jumps by more than five standard deviations is marked as suspect but never deleted. Analysts can override any flag, and every override is written to an audit log with a short reason.

4 Nightly checks

Every night a batch job compares each station with its two nearest neighbours. Large and lasting differences usually point to fouling or a loose cable rather than a real event. The job sends a short summary to the field team before seven in the morning.

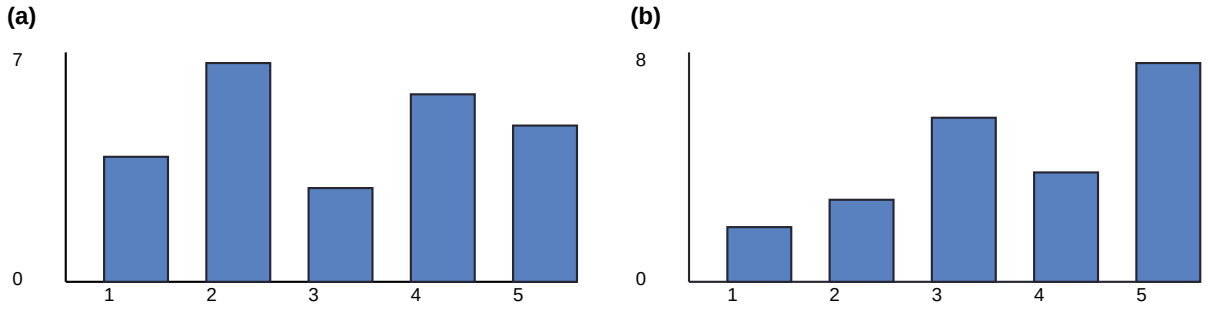

Figure 2: Faults per week during the first winter. Panel a counts fouling events and panel b counts link outages.

5 Archive

The archive keeps every summary and every raw sample in daily files, one file per station. Files are compressed after thirty days and copied to a second site once a week. A small web page lists the latest summary of each station and the time it arrived. Researchers can request the raw files through a form that checks their affiliation.

Table 2: Storage used by the archive after the first year.

Data	Files	Size
Summaries	8760	1.2 GB
Raw samples	8760	96 GB
Audit log	365	40 MB

The second phase of the project will add dissolved oxygen sensors to six of the stations. We also plan to replace the oldest batteries before the next winter season starts. Comments on this report are welcome and should be sent to the network coordinator.